ECA56 – ADVANCED WIRELESS COMMUNICATION SYSTEMS

ASSIGNMENT – 1

Cellular Network Evolution, Cellular Planning, Spectrum Allocation

Item	Details
Course	ECA56 – Advanced Wireless Communication Systems
Assignment	Assignment – 1
Topic	Cellular Network Evolution, Cellular Planning, Spectrum Allocation
Assessment	Open-ended Engineering Design Report
Maximum Marks	100
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1. Introduction

The proposed project is a state-wide wireless network intended to provide a common 5G/6G-ready communication platform across six states. The network must serve very different locations, from dense metropolitan areas to villages, highways, industrial zones and healthcare centres. Hence, the design cannot depend on a single frequency band or a single type of cell.

The alternative design presented here uses a coverage-first planning philosophy. A reliable low-frequency layer is established first, followed by a mid-band capacity layer. High-frequency mmWave cells are introduced only at locations where traffic demand is sufficiently high. This approach reduces unnecessary deployment cost while maintaining a path toward future 6G services.

The design also combines network slicing, edge computing, AI-based resource management, mobility optimization, cybersecurity and energy-saving techniques.

2. Stage 1 – Engineering Requirements

2.1 Regional Requirement Analysis

State	User/environment profile	Important requirement	Planning approach
Tamil Nadu	Cities, factories, ports, highways	Industrial reliability + broadband	Mid-band dense layer with low-band coverage and selected hotspots
Karnataka	Technology centres, cities, highways	High data traffic + low latency	Mid-band capacity, edge computing and small cells
Kerala	Dense settlements, coastal/hilly areas	Coverage + healthcare + resilience	Low-band coverage with strategically placed mid-band cells
Maharashtra	Major metropolitan and industrial regions	Very high capacity	Dense RAN, small cells, edge and selective mmWave
Gujarat	Industrial corridors, ports and cities	Industrial IoT + logistics	Mid-band industrial coverage and edge processing
Uttar Pradesh	Large rural and urban population	Wide coverage + affordable capacity	Low-band macro network with targeted mid-band expansion

2.2 Service Requirements

Service	Category	Required characteristics	Proposed solution
Mobile broadband	eMBB	High throughput	Mid-band, MIMO, small cells and mmWave hotspots
Autonomous vehicles	URLLC	Low delay and reliable connectivity	Edge UPF, reserved resources and overlapping coverage
Remote healthcare	URLLC/eMBB	Reliable video and low delay	Dedicated slice + edge site + redundant backhaul
Factory automation	URLLC/mMTC	Reliable control and many devices	Industrial small cells + edge computing
Smart meters/sensors	mMTC	Large device population	Wide-area low-band and efficient device scheduling
Emergency services	URLLC / public safety	High availability and priority	Dedicated slice and traffic pre-emption

2.3 Main Constraints

- Geographical diversity makes uniform cell planning unsuitable.
- Low frequencies provide coverage but normally have less bandwidth available for very high capacity.
- High-frequency deployment can deliver large capacity but requires dense sites because of shorter practical range and blockage.
- Adding cells increases capacity but also increases site, power, backhaul and maintenance requirements.
- More aggressive spectrum reuse can increase capacity while increasing interference-management requirements.
- Critical applications need better reliability than ordinary best-effort broadband.
- Security must be maintained across radio, transport, core, edge and application interfaces.
- Energy consumption must be controlled because the network will contain many radio and computing elements.

2.4 Problem Formulation

Let the design variables include the number and locations of macro/small cells, frequency assignment, transmit power, antenna configuration, spectrum allocation, edge-node placement and slice resource allocation.

The optimization objective is to minimize total deployment cost, energy use, interference and latency while maximizing coverage, capacity, reliability and spectral efficiency.

Subject to the constraints of available spectrum, site availability, power limits, required coverage, service latency, reliability targets and a blocking probability of not more than 2%.

2.5 Performance Measures

Parameter	Desired engineering objective
Coverage	High coverage probability in populated and critical corridors
Throughput	Adequate capacity during busy hours
Latency	Low end-to-end delay for critical services
Reliability	High availability for URLLC and public safety
Blocking probability	≤ 2% for the specified traffic model
Spectral efficiency	Increase bits/s/Hz through MIMO and beamforming
Cell utilization	Avoid sustained overload
Energy	Reduce energy per delivered bit
Handover	Low handover failure and interruption

Design Decision 1

The design prioritizes reliability and coverage first for essential services, followed by capacity, latency, scalability and cost. Broadband traffic is allowed to use flexible resources, while emergency and URLLC traffic receives protected resources.

3. Stage 2 – Spectrum and Cellular Planning

3.1 Frequency-Band Evaluation

Band	Strength	Limitation	Preferred deployment
Sub-1 GHz	Large coverage area and useful penetration	Limited bandwidth for extreme capacity	Rural areas, highways, IoT and emergency coverage
1–6 GHz	Balanced coverage and capacity	Less range than low-band	Main nationwide 5G capacity layer
>24 GHz mmWave	Very large bandwidth	Shorter range and blockage sensitivity	Dense hotspots, factories, stadiums and fixed high-rate links

The proposed spectrum strategy is therefore not a single-band design. Low-band is treated as the reliability and coverage foundation, mid-band as the workhorse capacity layer, and mmWave as an additional hotspot layer.

3.2 Cellular Planning Strategy

Cell	Purpose	Example location
Macro gNodeB	Wide-area service and mobility	Rural areas, highways, suburbs
Micro/small cell	Capacity enhancement	Urban streets and transport hubs
Enterprise/industrial small cell	Reliable local connectivity	Factories and industrial parks
mmWave small cell	Extreme local capacity	High-density hotspots
Edge node	Local processing	Cities, factories and transport corridors

3.3 Frequency Reuse

Frequency reuse permits the same spectrum to be used again in geographically separated cells. In a modern 5G system, reuse can be close to one because advanced antenna techniques and interference coordination are available. However, dense deployments require careful power control, beam selection and scheduling.

The design therefore uses spatial reuse rather than depending only on large frequency separation. This increases spectral efficiency but requires continuous interference monitoring.

3.4 Capacity Enhancement

- Cell splitting: introduce additional cells in overloaded regions.
- Sectoring: divide a cell into directional sectors to increase spatial reuse.
- Carrier aggregation: combine available carriers to increase user data rate.
- Massive MIMO: serve multiple users using spatial multiplexing.
- Beamforming: focus transmitted energy toward the intended user.
- Traffic steering: move users between low-band, mid-band and small-cell layers.

3.5 Illustrative Cell Calculation

Assume an urban planning zone produces 7.2 Gbps of busy-hour demand and one effective cell provides 600 Mbps under the selected operating conditions.

Minimum cells = 7.2 Gbps / 0.6 Gbps = 12 cells.

If 20% additional capacity is reserved for traffic growth and variation: Planned cells = ceil(12 × 1.20) = 15 cells.

Thus, 15 cells are planned for this illustrative zone. This is a capacity calculation only; the final count must also satisfy coverage and propagation requirements.

3.6 Interference Control

Problem	Control method
Co-channel interference	Beamforming, coordinated scheduling and power control
Adjacent-channel interference	Filtering and suitable channel separation
Dense small-cell interference	Adaptive power and interference-aware association
Cell-edge interference	Coordinated resource allocation and antenna optimization
mmWave blockage	Multiple beams, fallback to lower-frequency coverage

State-wise Radio Plan

State	Primary layer	Additional layer	Special focus
Tamil Nadu	Mid-band	Low-band + mmWave hotspots	Manufacturing, ports, cities
Karnataka	Mid-band	Small cells + edge	Technology centres and highways
Kerala	Low-band + mid-band	Selected small cells	Healthcare and resilience
Maharashtra	Mid-band	Dense small cells + mmWave	Very high urban traffic
Gujarat	Mid-band	Industrial small cells	Ports and industrial automation
Uttar Pradesh	Low-band	Mid-band city/corridor layer	Large-area economical coverage

Design Decision 2

A mixed radio architecture is recommended. A uniform mmWave network would be expensive and coverage-limited, while a low-band-only network would not provide sufficient capacity. The combination gives the best practical balance.

4. Stage 3 – Quantitative Models

4.1 Example Traffic Model

Traffic source	Active units	Assumed rate	Total
HD video	8,000 users	2.5 Mbps	20 Gbps
Autonomous transport	3,000 vehicles	0.8 Mbps	2.4 Gbps
Industrial IoT	120,000 devices	0.015 Mbps	1.8 Gbps

Illustrative aggregate demand = 20 + 2.4 + 1.8 = 24.2 Gbps.

With a 20% planning margin, required capacity = 24.2 × 1.20 = 29.04 Gbps.

The example shows why capacity must be distributed over many cells instead of being supplied by one large cell.

4.2 Bandwidth and Spectral Efficiency

The practical relationship is R = Bη. If an effective spectral efficiency of 4.5 bit/s/Hz is assumed, a 500 Mbps cell requires approximately B = 500/4.5 = 111.1 MHz of effective bandwidth. Since real systems have overhead and varying spectral efficiency, actual planning must use measured or simulated throughput rather than the ideal expression alone.

The Shannon expression $C = B \log_2(1+SNR)$ is useful as a theoretical upper limit. Actual 5G throughput is reduced by interference, protocol overhead, channel conditions and scheduling.

4.3 Propagation and Coverage

Propagation loss increases with distance and is affected by frequency, antenna characteristics and the environment. As frequency increases, practical blockage becomes more important. The proposed network therefore maintains a lower-frequency coverage layer beneath higher-frequency capacity layers.

Urban planning should consider building penetration and street-level blockage, while rural planning should consider terrain, tower height and large inter-site distances.

4.4 Latency Allocation

Network part	Illustrative contribution	Improvement
Radio access	5–8 ms	Efficient scheduling and suitable numerology
Backhaul/transport	5–8 ms	Short routes and resilient high-capacity links
Core	3–5 ms	Cloud-native functions and local UPF
Edge application	2–4 ms	Process close to users
Total	Approximately 15–25 ms	Application-specific validation required

The values are illustrative planning targets rather than guaranteed performance. URLLC applications require application-specific testing.

4.5 Erlang-B Example

The Erlang-B equation is used to estimate the probability that an offered traffic load is blocked when all channels are occupied:

B(E,N) = (E^N/N!) / [Σ(k=0 to N) E^k/k!].

For an alternative numerical example, assume 750 busy-hour subscribers and 0.025 Erlang per subscriber. Offered traffic E = 750 × 0.025 = 18.75 Erlangs.

Using the recursive Erlang-B calculation, N = 27 channels gives a blocking probability of approximately 1.60%, satisfying the 2% target.

This calculation intentionally treats the channels as ideal traffic channels. Guard bands, duplexing overhead, control resources and other practical spectrum overheads are excluded from this simplified channel-count calculation. The actual supplied/operator traffic values should replace the assumptions.

4.6 Reliability and Mobility

Autonomous vehicles require continuous service as they move between cells. Overlapping coverage, optimized neighbor lists, measurement-based handover and rapid target-cell preparation are required. A lower-frequency layer should remain available so that a vehicle does not lose service when a high-frequency beam is blocked.

For emergency services, reliability is increased by redundant backhaul, protected network slices and priority resource allocation.

4.7 Design Decision 3

State	Main expected bottleneck	Intervention
Tamil Nadu	Industrial and city capacity	Densification and edge computing
Karnataka	Peak urban traffic	Small cells and traffic steering
Kerala	Coverage/resilience	Low-band expansion and redundant links
Maharashtra	Very high capacity	Dense small cells and selective mmWave
Gujarat	Industrial reliability	Industrial RAN and edge
Uttar Pradesh	Coverage over large area	Low-band macro expansion and targeted mid-band

5. Stage 4 – Architecture Evaluation

5.1 Alternative 1: Coverage-Oriented Macro Network

Alternative 1 uses mostly macro cells with low-band and mid-band spectrum. Its major advantage is simpler deployment with fewer sites. It is suitable for broad rural coverage but is less effective for very dense traffic and short-latency enterprise applications.

5.2 Alternative 2: Layered Heterogeneous Network

Alternative 2 uses macro cells, small cells, mid-band spectrum, selective mmWave, edge computing and service-aware network slicing. It requires more planning and investment but supports a much wider range of applications.

5.3 Decision Matrix

Criterion	Weight	Alternative 1	Alternative 2
Coverage	20	5	4
Capacity	20	3	5
Latency	15	3	5
Reliability	15	4	5
Scalability	10	3	5
Complexity	5	5	3
Energy	5	4	4
Economic efficiency	10	5	3
Weighted result	100	3.90/5	4.50/5

Alternative 2 is selected because the assignment requires support for high-speed broadband, autonomous transport, healthcare, industrial automation, IoT and emergency communication at the same time.

5.4 Network Slices

Slice	Users	Main policy
eMBB	Broadband/video users	Maximize throughput
URLLC	Vehicles, industry, healthcare	Reserve low-latency resources
mMTC	Sensors and meters	Support large device numbers efficiently
Public safety	Emergency organizations	Highest priority during incidents

5.5 Edge Placement

Edge resources are recommended near major cities, industrial parks, transport corridors and healthcare clusters. Local UPF and application processing can reduce the distance that critical traffic travels before reaching its application.

Design Decision 4

The heterogeneous design remains the preferred architecture. If the objective changes to rural coverage only, more low-band macro cells and fewer small cells would be selected. If vehicle reliability becomes dominant, more edge nodes, redundant links and overlapping transport-corridor coverage would be added.

6. Stage 5 – Integrated Network Design

6.1 Proposed Architecture

User equipment and IoT devices connect to gNodeBs through the selected low-band, mid-band or mmWave layer. The RAN connects through redundant backhaul to the 5G Core. AMF provides access and mobility management, SMF manages sessions, and UPF carries user traffic. Local UPF and edge application servers are used for latency-sensitive traffic.

Logical flow:

Users / Vehicles / IoT → gNodeB → 5G RAN → Redundant Backhaul → 5G Core → UPF / Edge → Applications / Cloud

6.2 Integrated Control Functions

Function	Role
Radio resource management	Assign time-frequency resources according to traffic and radio conditions
Mobility management	Maintain service during cell changes
Slice management	Reserve and prioritize resources by service
AI controller	Predict traffic and recommend optimization actions
Security controller	Monitor authentication, anomalies and incidents
Energy controller	Reduce unnecessary radio and computing power

6.3 AI-Based Optimization

The AI system receives cell utilization, user counts, signal quality, traffic history and mobility information. It predicts congestion and recommends resource allocation before the cell becomes overloaded.

- Predict traffic load for the next planning interval.
- Identify cells approaching congestion.
- Shift users to available neighboring layers.
- Adjust radio resources and transmit power.
- Activate or deactivate suitable small-cell capacity.
- Compare the result with a conventional rule-based baseline.

6.4 Security and Resilience

- Strong authentication should be applied to users and network functions.
- Control and user-plane traffic should be protected using appropriate encryption.
- Anomaly detection should monitor unusual signaling and traffic behaviour.
- Critical network functions should have redundancy where economically justified.
- Backhaul should use alternate paths so a single link failure does not isolate a critical region.
- Emergency traffic should be allowed priority access through the public-safety slice.
- Recovery procedures should define how services are restored after equipment or transport failure.

6.5 Energy Efficiency

- Use low-power operation or sleep modes for lightly loaded small cells when service permits.
- Use traffic-aware resource allocation to avoid unnecessary active resources.
- Share towers, sites, power and transport infrastructure where possible.

- Use efficient cooling and power management for edge sites.
- Monitor energy per delivered bit as a long-term KPI.

6.6 6G Evolution

The network should reserve an evolution path for AI-native networking and Integrated Sensing and Communication (ISAC). These concepts can be evaluated through research and pilot deployments. Future THz communication may support extremely high data rates at short distances, but it should not be treated as a current nationwide access layer.

Design Decision 5

The final design intentionally separates current 5G functions from future 6G concepts. 5G NR, gNodeBs, 5G Core, massive MIMO, beamforming, slicing and edge computing form the practical base. AI-native networking, ISAC and future THz systems are treated as evolutionary research directions.

7. Stage 6 – Validation

7.1 Validation Plan

Parameter	Validation technique	Result to observe
Traffic	Analytical model/Python	Peak demand and capacity margin
Blocking	Erlang-B	Probability compared with 2% target
Coverage	Link budget / propagation model	Coverage probability and dead zones
Throughput	MATLAB/ns-3/system model	User and cell throughput
Latency	End-to-end model	Application-specific delay
Handover	Mobility simulation	Handover success and interruption
Interference	Radio simulation	SINR and throughput degradation
Energy	Power model	Energy per delivered bit

7.2 Peak Traffic Scenario

If video traffic increases sharply during a busy-hour event, the AI controller identifies overloaded cells and steers users to nearby small cells or alternate carriers. URLLC and public-safety slices remain protected. If overload persists, temporary capacity can be added or additional spectrum activated.

7.3 Infrastructure Failure Scenario

If a backhaul link fails, routing should move critical traffic to an alternate path. If an edge node fails, applications should use a redundant edge node or controlled regional-cloud fallback. During an emergency, ordinary broadband traffic may be deprioritized so that public-safety traffic continues.

7.4 Verification Table

Requirement	Design response	Verification
Wide-area coverage	Low-band macro layer	Coverage analysis
High urban capacity	Mid-band + small cells	Throughput/utilization analysis
Hotspot capacity	mmWave	Hotspot simulation
2% blocking	Erlang-B sizing	Traffic calculation
Low latency	Edge + URLLC slice	Latency analysis
Mobility	Overlapping cells + handover	Mobility simulation
Massive IoT	mMTC slice	Device-load evaluation
Emergency reliability	Priority slice + redundancy	Failure scenario
AI optimization	Prediction + dynamic RRM	Baseline comparison
Sustainability	Sleep modes + sharing	Energy model

8. Economic, Environmental and Operational Evaluation

Factor	Positive effect	Trade-off
Economic	Layered deployment avoids universal mmWave cost	Small-cell and edge deployment increases capital cost
Operational	AI and automation reduce manual optimization effort	Requires skilled monitoring and model management
Environmental	Energy-aware operation reduces power use	More sites may increase total equipment footprint
Public safety	Dedicated resources improve emergency availability	Redundant infrastructure increases cost
Scalability	Cloud-native core and slices support new services	More software orchestration is required

9. Limitations and Further Data Required

- The question paper does not provide numerical subscriber counts, busy-hour traffic per subscriber or actual channel counts; therefore, the numerical examples are stated assumptions.
- Actual spectrum availability and licensing conditions must be confirmed before implementation.
- Site surveys are required to determine tower locations, antenna heights, terrain and building losses.
- Measured traffic traces are required to replace the illustrative traffic model.
- Real propagation and system simulations should be used before final cell-count decisions.

- Security and regulatory compliance must be verified against the applicable operational environment.

10. Final Recommendation

The recommended solution is a phased heterogeneous 5G/6G-ready network. The first layer should provide reliable low-band coverage across rural regions, highways and critical public-safety areas. The second layer should provide the majority of capacity through mid-band macro and small cells. The third layer should use mmWave only at locations where traffic demand justifies it.

The network should use a 5G Core with network slicing and local edge processing. Four logical slices should support eMBB, URLLC, mMTC and public safety. AI-based resource management should be introduced after collecting adequate network measurements, and its performance should be compared with a baseline method.

The recommended deployment is therefore 'deployment with modifications and validation'. The design is technically promising, but actual implementation requires operator-specific spectrum, traffic, propagation, site and cost data.

11. Conclusion

A six-state communication system cannot be optimized effectively with one spectrum band or one cell type. The alternative design uses a layered approach so that each frequency range performs the role for which it is most suitable. Low-band provides wide coverage, mid-band supplies the main capacity, and mmWave adds localized high-throughput capability.

The combination of OFDMA, massive MIMO, beamforming, frequency reuse, small-cell densification, network slicing and edge computing supports the diverse requirements of broadband, autonomous transport, healthcare, industrial automation, IoT and emergency services. AI-based optimization can further improve traffic management and energy efficiency.

The quantitative models demonstrate how traffic demand, spectral efficiency, cell sizing, latency and Erlang-B blocking can support engineering decisions. Final deployment should replace all illustrative values with measured network data and should be validated using MATLAB, Python, ns-3 or equivalent tools.